

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	18 June 2026
Team ID	
Project Name	SHOPEZ : E-commerce Application
Maximum Marks	4 Marks

Technical Architecture:

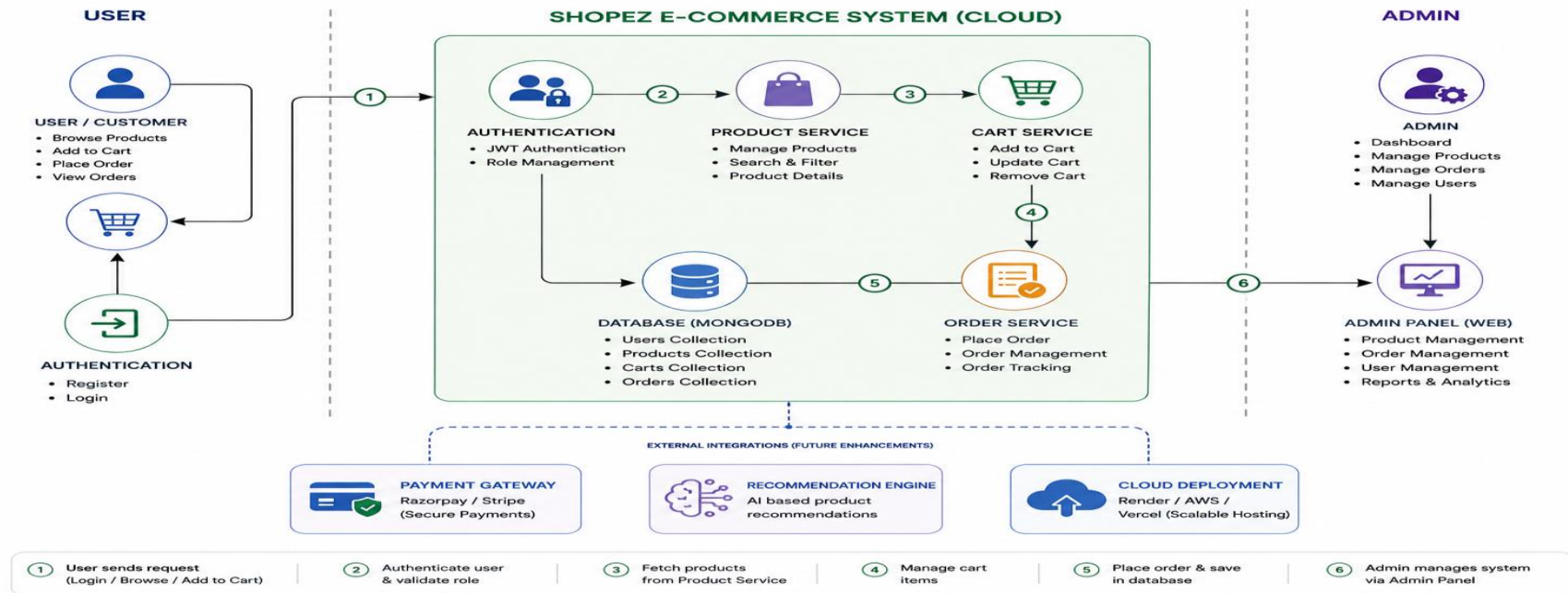


Table-1 : Components

S.No	Component	Description	Technology
1	User Interface	Web application for customers and administrators	React.js, HTML5, CSS3, JavaScript
2	Application Logic-1	User Authentication and Authorization	Node.js, Express.js, JWT
3	Application Logic-2	Product Management, Search, Filter and Cart Operations	Node.js, Express.js
4	Application Logic-3	Order Processing and Admin Operations	Node.js, Express.js
5	Database	Stores users, products, carts and orders	MongoDB (NoSQL)
6	Cloud Database	Cloud-hosted database service	MongoDB Atlas
7	File Storage	Product images and application files	Local File System / Cloud Storage
8	External API-1	Payment Processing (Future Enhancement)	Razorpay / Stripe API
9	External API-2	Email Notifications & Verification	Nodemailer / Gmail SMTP
10	Machine Learning Model	Product Recommendation System (Future Enhancement)	Recommendation Engine using Python ML
11	Infrastructure (Server / Cloud)	Application Deployment	Local Server, Render, Vercel, AWS

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Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used in development	React.js, Node.js, Express.js, MongoDB
2	Security Implementations	Authentication, authorization and password protection	JWT Authentication, Bcrypt Password Hashing, HTTPS, OWASP Security Practices
3	Scalable Architecture	Three-tier architecture separating frontend, backend and database	React.js + Express.js + MongoDB
4	Availability	Cloud deployment with high availability and backup support	MongoDB Atlas, Render/Vercel
5	Performance	Fast API responses, optimized database queries and caching support	MongoDB Indexing, REST APIs, React Virtual DOM

